

Chapter 1

Electron Capture in a $v = c$ Travelling-Wave Linac

1.1 Introduction

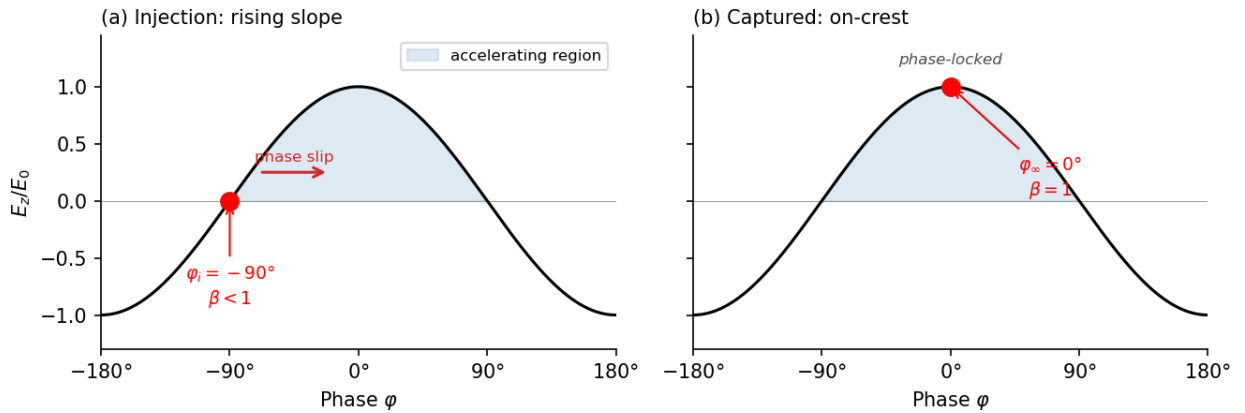
For relativistic electrons ($\beta \approx 1$), disc-loaded waveguide structures are designed with phase velocity $v_{\text{ph}} = c$. A particle injected with $\beta < 1$ is slower than the wave and slips in phase, but accelerates as it gains energy. If the field is strong enough the particle reaches $\beta \approx 1$ before slipping past the accelerating region, and thereafter rides the wave at a fixed phase.

Because no synchronous particle exists during the capture transient, the relevant variable is the **absolute** phase φ (not a deviation $\Delta\varphi$).

1.2 Equations of motion

Define the wave phase at the particle position s :

$$\varphi(s, t) = \omega t - \frac{2\pi}{\lambda} s.$$



1.2.1 Phase slip

The particle velocity is $ds/dt = \beta c$, while the wave travels at c . Differentiating φ with respect to time:

$$\frac{d\varphi}{dt} = \omega - \frac{2\pi}{\lambda}\beta c = \frac{2\pi c}{\lambda}(1 - \beta).$$

Converting to the independent variable s using $dt = ds/(\beta c)$:

$$\boxed{\frac{d\varphi}{ds} = \frac{2\pi}{\lambda} \frac{1 - \beta}{\beta}}. \quad (1)$$

Since $\beta < 1$, $d\varphi/ds > 0$: the particle continually slips to later phase.

1.2.2 Energy gain

The on-axis accelerating field of the travelling wave is $\mathcal{E}_z = E_0 \cos \varphi$. The energy $W = \gamma m_0 c^2$ changes as:

$$\frac{dW}{ds} = qE_0 \cos \varphi \quad \Longrightarrow \quad \boxed{m_0 c^2 \frac{d\gamma}{ds} = qE_0 \cos \varphi}. \quad (2)$$

1.3 Reduction to a single ODE

Dividing Eq. (2) by Eq. (1) eliminates s :

$$\frac{d\gamma}{d\varphi} = \frac{qE_0 \lambda}{2\pi m_0 c^2} \frac{\beta \cos \varphi}{1 - \beta}. \quad (3)$$

We convert to a $d\beta/d\varphi$ form using the identity $d\gamma = \gamma^3 \beta d\beta$ (which follows from $\gamma = (1 - \beta^2)^{-1/2}$):

$$\gamma^3 \beta \frac{d\beta}{d\varphi} = \frac{qE_0 \lambda}{2\pi m_0 c^2} \frac{\beta \cos \varphi}{1 - \beta}.$$

The factor β cancels. Writing $\gamma^3 = (1 - \beta^2)^{-3/2}$:

$$\frac{1}{(1 - \beta^2)^{3/2}} \frac{d\beta}{d\varphi} = \frac{qE_0 \lambda}{2\pi m_0 c^2} \frac{\cos \varphi}{1 - \beta}.$$

Multiplying both sides by $(1 - \beta)$ and noting $(1 - \beta^2)^{3/2} = (1 - \beta)^{3/2}(1 + \beta)^{3/2}$:

$$\boxed{\frac{1}{(1 - \beta)^{1/2}(1 + \beta)^{3/2}} \frac{d\beta}{d\varphi} = \frac{qE_0 \lambda}{2\pi m_0 c^2} \cos \varphi}. \quad (4)$$

1.4 Integration via the substitution $\beta = \cos \zeta$

The LHS of Eq. (4) can be integrated in closed form with the substitution $\beta = \cos \zeta$ ($d\beta = -\sin \zeta d\zeta$). The half-angle identities give:

$$1 - \beta = 2 \sin^2 \frac{\zeta}{2}, \quad 1 + \beta = 2 \cos^2 \frac{\zeta}{2}, \quad \sin \zeta = 2 \sin \frac{\zeta}{2} \cos \frac{\zeta}{2}.$$

Substituting into the LHS integrand:

$$\begin{aligned} \frac{d\beta}{(1 - \beta)^{1/2}(1 + \beta)^{3/2}} &= \frac{-\sin \zeta d\zeta}{(\sqrt{2} \sin \frac{\zeta}{2})(2 \cos^2 \frac{\zeta}{2})^{3/2}} \\ &= \frac{-2 \sin \frac{\zeta}{2} \cos \frac{\zeta}{2} d\zeta}{\sqrt{2} \sin \frac{\zeta}{2} \cdot 2\sqrt{2} \cos^3 \frac{\zeta}{2}} = \frac{-d\zeta}{2 \cos^2 \frac{\zeta}{2}}. \end{aligned}$$

Integrating:

$$\int \frac{d\beta}{(1-\beta)^{1/2}(1+\beta)^{3/2}} = -\tan\frac{\zeta}{2} + C.$$

Expressing the result in terms of β via

$$\tan\frac{\zeta}{2} = \sqrt{\frac{1-\cos\zeta}{1+\cos\zeta}} = \sqrt{\frac{1-\beta}{1+\beta}},$$

and integrating the RHS of Eq. (4) from (β_i, φ_i) to (β, φ) :

$$-\sqrt{\frac{1-\beta}{1+\beta}} + \sqrt{\frac{1-\beta_i}{1+\beta_i}} = \frac{qE_0\lambda}{2\pi m_0 c^2} (\sin\varphi - \sin\varphi_i).$$

Solving for $\sin\varphi$:

$$\boxed{\sin\varphi = \sin\varphi_i + \frac{2\pi m_0 c^2}{qE_0\lambda} \left[\sqrt{\frac{1-\beta_i}{1+\beta_i}} - \sqrt{\frac{1-\beta}{1+\beta}} \right].} \quad (5)$$

This is the exact phase-velocity relation during capture (Wangler eq. 6.70).

1.5 Asymptotic phase

Because the particle accelerates ($\beta > \beta_i$), the second term on the RHS of Eq. (5) is positive and increases with β . Therefore $\sin\varphi > \sin\varphi_i$, and the phase becomes more positive as β increases. If we want the particle to approach the crest where $\varphi = 0$, it must be injected at a negative phase.

As $\beta \rightarrow 1$ the square root $\sqrt{(1-\beta)/(1+\beta)} \rightarrow 0$, and the particle settles at a final phase φ_∞ :

$$\boxed{\sin\varphi_\infty = \sin\varphi_i + \frac{2\pi m_0 c^2}{qE_0\lambda} \sqrt{\frac{1-\beta_i}{1+\beta_i}}.} \quad (6)$$

This is the central result of the capture analysis (Wangler eq. 6.71).

1.5.1 The capture parameter F

Define

$$F \equiv \frac{2\pi m_0 c^2}{qE_0\lambda} \sqrt{\frac{1-\beta_i}{1+\beta_i}}. \quad (7)$$

Then $\sin\varphi_\infty = \sin\varphi_i + F$ (Wangler eq. 6.72).

1.5.2 On-crest capture

Choosing $\varphi_i \approx -\pi/2$ (or slightly more positive to ensure acceleration) and $F = 1$ places the asymptotic phase at the crest, giving:

$$\boxed{E_0 = \frac{2\pi m_0 c^2}{q\lambda} \sqrt{\frac{1-\beta_i}{1+\beta_i}}.} \quad (8)$$

This shows that the smaller the initial velocity, the larger the accelerating field must be (Wangler eq. 6.73).

1.6 Phase compression during capture

The capture equation $\sin \varphi_\infty = \sin \varphi_i + F$ maps a range of injection phases to a narrower range of asymptotic phases. Consider particles distributed about $\varphi_i = -\pi/2$ with spread $\pm\delta$, and set $F = 1$ (on-crest design).

For a particle at $\varphi_i = -\pi/2 + \delta$:

$$\sin \varphi_\infty = \sin(-\frac{\pi}{2} + \delta) + 1 = -\cos \delta + 1 \approx \frac{\delta^2}{2}.$$

Since φ_∞ is small, $\sin \varphi_\infty \approx \varphi_\infty$, giving

$$\boxed{\varphi_\infty \approx \frac{\delta^2}{2}}. \quad (9)$$

Both $+\delta$ and $-\delta$ map to the same $\varphi_\infty \approx \delta^2/2$. An initial phase spread of 2δ is compressed to a final spread of order $\delta^2/2$ — a quadratic compression (SY Lee, Sec. VIII.4.A).

1.6.1 Example: $F = 1$

With $F = 1$ and initial phase length $2\delta = 20^\circ$ ($\delta = \pi/18$): - All particles within $\varphi_i \in (-\pi/2 - \delta, -\pi/2 + \delta)$ are captured. - The asymptotic phases lie in $\varphi_\infty \in [0, \delta^2/2] = [0, 0.015 \text{ rad}] \approx [0^\circ, 0.9^\circ]$. - A 20° injection spread compresses to $\lesssim 1^\circ$ at the crest.

1.6.2 Example: $F = 1.5$

A larger F (stronger field) captures particles further from $-\pi/2$. With $F = 1.5$ and $\sin \varphi_\infty = \sin \varphi_i + 1.5 \leq 1$: - The capture condition $\sin \varphi_i \leq -0.5$ gives $\varphi_i \in (-5\pi/6, -\pi/6)$. - Particles starting at $\varphi_i = -\pi/6$ reach $\sin \varphi_\infty = -0.5 + 1.5 = 1$, i.e. $\varphi_\infty = \pi/2$. - Particles starting at $\varphi_i = -\pi/2$ reach $\sin \varphi_\infty = -1 + 1.5 = 0.5$, i.e. $\varphi_\infty = \pi/6$. - The captured beam occupies $\varphi_\infty \in (\pi/6, \pi/2)$, which is past the crest — these particles are in the decelerating/focusing region.

The $F = 1$ case is therefore preferred for maximum energy gain (all captured particles near the crest), while $F > 1$ captures a wider phase range at the cost of placing particles off-crest.

1.6.3 Example: Phase freezing

Check fig. 1.1.

Parameter	Value
RF frequency	1300 MHz
Accelerating gradient E_0	10 MV/m
Gap spacing L	$\lambda/20 \approx 11.5 \text{ mm}$
Voltage per gap	115 kV
Injection energy	200 keV ($\beta_0 = 0.695$, $\gamma_0 = 1.39$)
Injection phases	$-60^\circ, -30^\circ, 0^\circ$
Number of gaps	150

The asymptotic (frozen) phase is a nonlinear function of the injection phase. Particles injected ahead of crest ($\varphi < 0$) slip forward in phase and freeze at a value closer to crest. We scan injection phases to find which one gives $\varphi_{\text{asympt}} = 0^\circ$ — this is the optimal injection phase for on-crest capture, check Fig. 1.2.

Parameter	Value
RF frequency	1300 MHz
Accelerating gradient E_0	10 MV/m

Parameter	Value
Gap spacing L	11.5 mm
Injection energy	200 keV
Injection phase scan	-90° to 0° (500 points)
Number of gaps (convergence)	300

1.6.4 Example: Phase slip

We inject a beam of 500 electrons at the optimal injection phase ($\varphi_{\text{opt}} \approx -39.7^\circ$) found above, with small spreads in phase and energy, check Fig. 1.3.

Parameter	Value
RF frequency	1300 MHz
Accelerating gradient E_0	10 MV/m
Gap spacing L	11.5 mm
Number of particles	500
Injection phase (centroid)	$\varphi_{\text{opt}} \approx -39.7^\circ$
Injection energy (centroid)	200 keV
Phase spread σ_φ	5°
Energy spread σ_W	10 keV
Number of gaps	120

1.6.5 Example: Proton vs electron

To highlight the difference between electron and proton dynamics, we track both species starting at the same injection phase. The proton undergoes **synchrotron oscillation** about the synchronous phase, while the electron **freezes** almost immediately, check Fig. 1.4.

Parameter	Electron	Proton
RF frequency	1300 MHz	400 MHz
Accelerating gradient E_0	10 MV/m	4 MV/m
Injection energy	200 keV	300 keV
Synchronous phase φ_s	N/A ($v_p = c$)	-30°
Gap length	$\lambda/20 = 11.5$ mm	$\beta\lambda = 19.0$ mm
Injection phase	$\varphi_{\text{opt}} \approx -39.7^\circ$	same
Number of gaps	200	200

1.7 Sine and cosine conventions

The main body of these notes (Wangler) uses the **cosine convention**: the travelling-wave field is $E_z = E_0 \cos \varphi$, so the crest sits at $\varphi = 0$. The standing-wave section (SY Lee) uses the **sine convention**: $E_z = E_0 \sin \varphi$, placing the crest at $\varphi = 90^\circ$. The two are related by a shift of time origin:

$$\varphi_{\text{sin}} = \varphi_{\text{cos}} + 90^\circ.$$

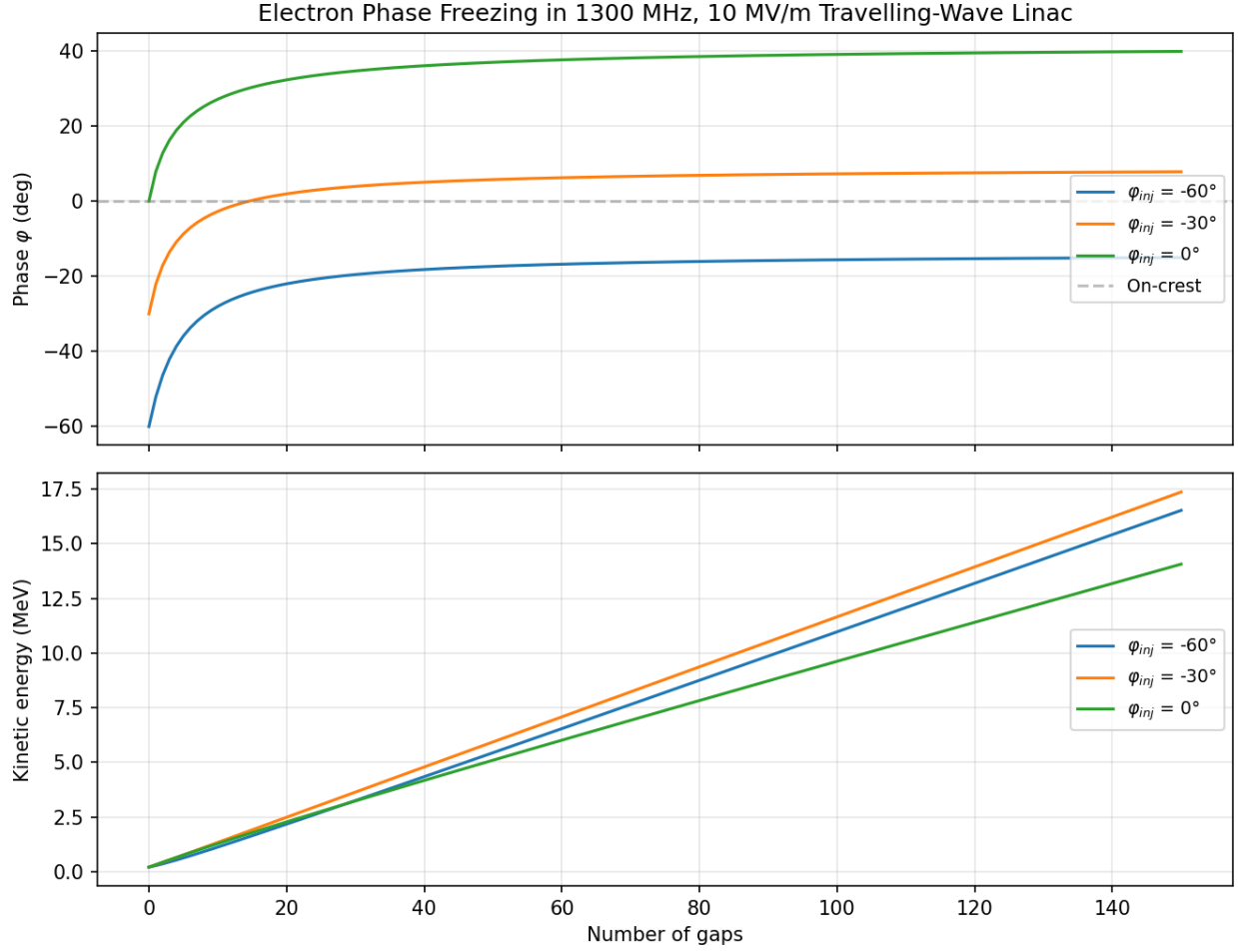


Figure 1.1: Phase freezing

Quantity	Cosine convention (Wangler)	Sine convention (SY Lee)
Field	$E_z = E_0 \cos \varphi$	$E_z = E_0 \sin \varphi$
Crest (max acceleration)	$\varphi = 0^\circ$	$\varphi = 90^\circ$
Zero-crossing (rising)	$\varphi = -90^\circ$	$\varphi = 0^\circ$
Phase mapping	$\varphi_{\cos}$	$\varphi_{\sin} = \varphi_{\cos} + 90^\circ$
TW capture (exact)	$\sin \varphi_\infty = \sin \varphi_i + F$	$\cos \varphi_\infty = \cos \varphi_i - F$
TW capture parameter	$F = \frac{2\pi m_0 c^2}{q E_0 \lambda} \sqrt{\frac{1 - \beta_i}{1 + \beta_i}}$	same
Optimal TW injection ($F = 1$, on-crest asymptote)	$\varphi_i = -90^\circ$ (zero-crossing)	$\varphi_i = 0^\circ$ (zero-crossing)

Both conventions describe identical physics — the particle sees the same field at the same instant. The difference is purely a choice of time origin (which field zero-crossing defines $\varphi = 0$). When reading the literature, check whether the author’s “accelerating crest” sits at $\varphi = 0$ (cosine/Wangler) or $\varphi = 90^\circ$ (sine/SY Lee).

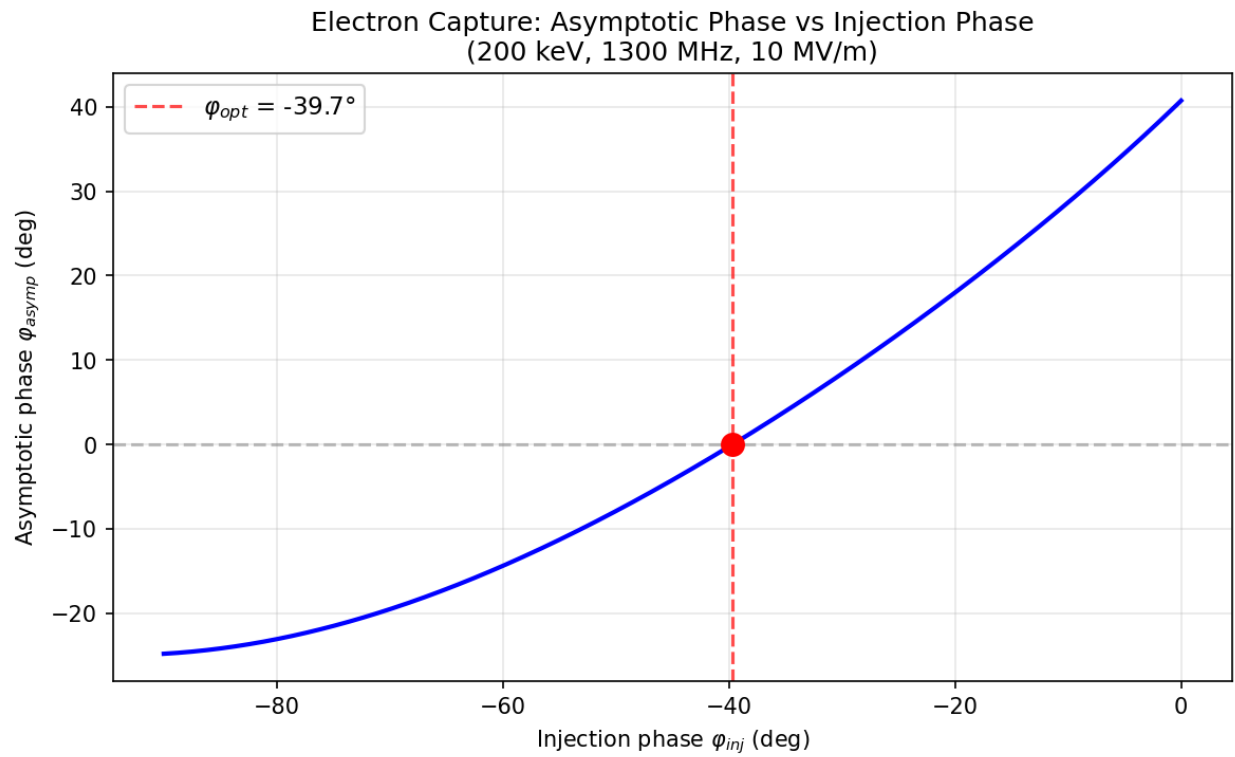


Figure 1.2: Asymptotic phase

1.8 References

- T. P. Wangler, *RF Linear Accelerators*, 2nd ed. (Wiley, 2008), Sec. 6.10.
- S. Y. Lee, *Accelerator Physics*, 4th ed. (World Scientific), Sec. VIII.4.A.

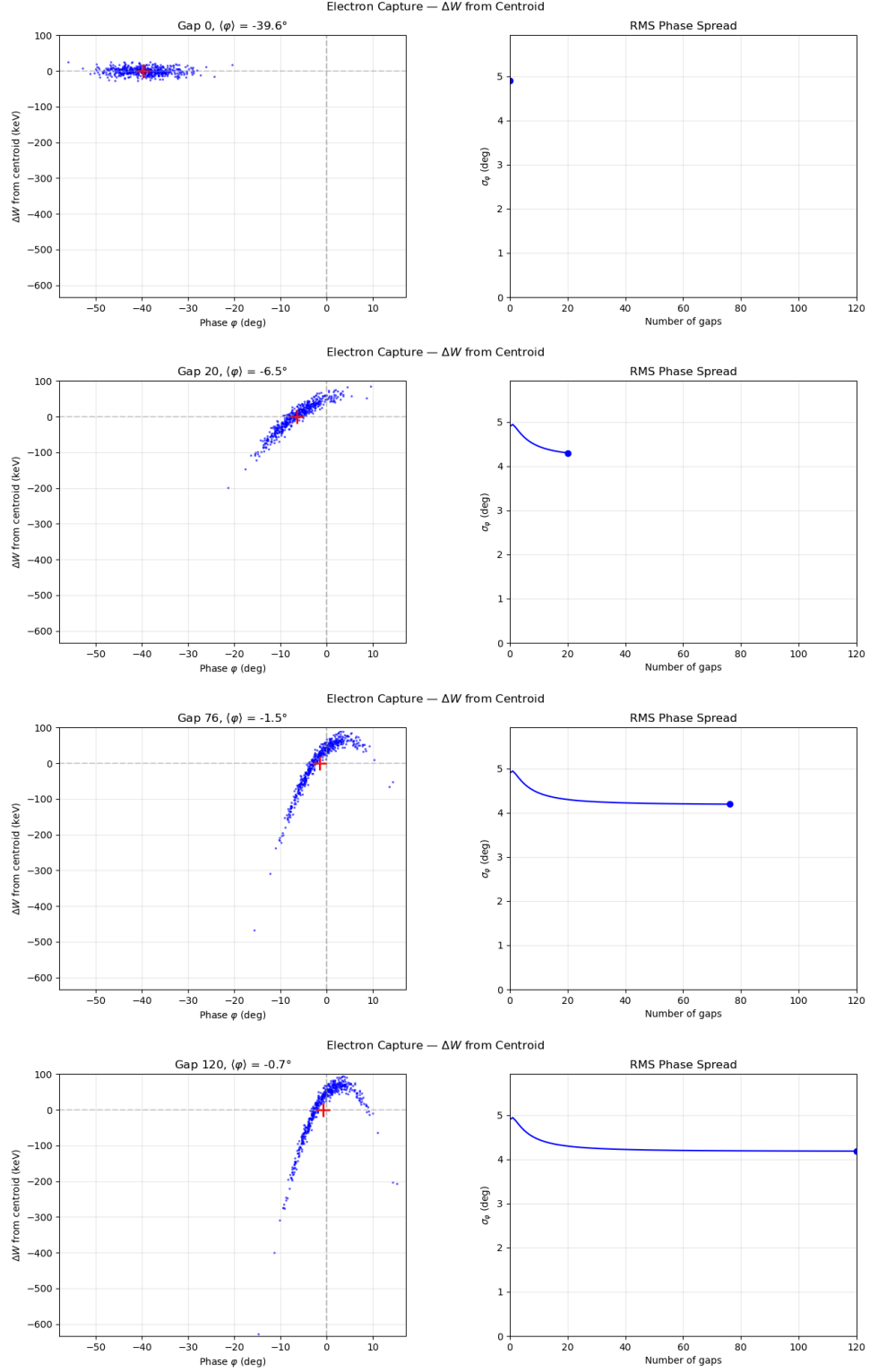


Figure 1.3: Phase space evolution

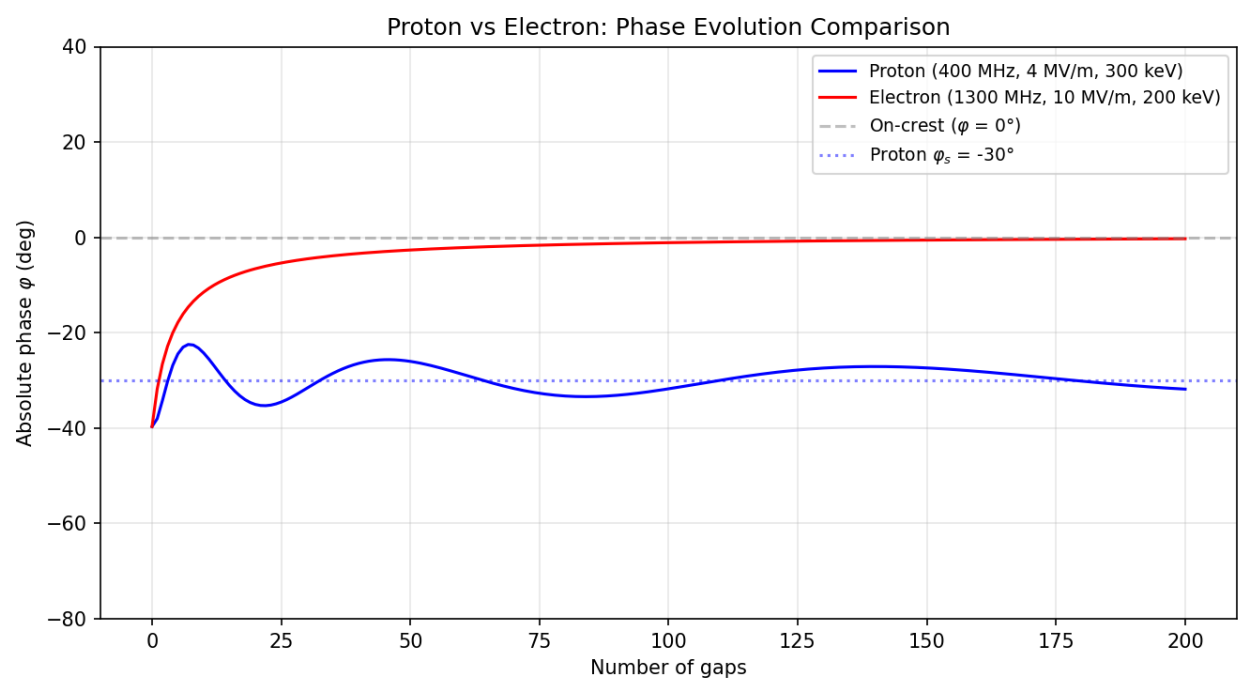


Figure 1.4: Proton vs electron

Chapter 2

Magnetic Bunch Compression for Electron Beams

2.1 Introduction

Free-electron lasers (FELs) require electron bunches with peak currents of order kA, yet photocathode guns produce bunches with peak currents of only tens of amperes. The bunch must therefore be compressed longitudinally by a factor of 10–100 between the gun and the undulator.

The compression strategy is:

1. Impose an **energy–position correlation** (chirp) on the bunch using an RF linac operated off-crest.
2. Pass the chirped bunch through a **dispersive transport** (magnetic chicane) where higher-energy particles travel a shorter path, allowing the tail to catch up with the head.

2.2 Longitudinal phase-space coordinates

For relativistic electron beams ($\beta \approx 1$), the natural longitudinal coordinates are (z, δ) :

$$z = c \Delta t, \quad \delta = \frac{E - E_s}{E_s},$$

where $\Delta t = t - t_s$ is the arrival-time difference with respect to the reference particle, and E_s is the reference energy.

Sign convention (ELEGANT): A particle *ahead* of the reference arrives *earlier* ($\Delta t < 0$), so:

$z = c \Delta t < 0 \quad \text{for a particle at the bunch head.}$

2.3 Longitudinal tracking equations

For beam transport through linac sections and chicanes, we use thin-lens maps rather than differential equations.

2.3.1 RF acceleration (thin-lens kick)

A relativistic particle traversing an RF cavity of voltage V and wavenumber $k = 2\pi/\lambda$ acquires energy:

$$\boxed{\begin{aligned} z_f &= z_i, \\ E_f &= E_i + eV \cos(k z_i + \varphi_s), \end{aligned}} \quad (1)$$

where φ_s is the synchronous phase. The position z is unchanged (thin-lens: no path-length change inside the cavity).

For compression, the linac is operated at $\varphi_s \neq 0$ (off-crest) to imprint an energy chirp: particles at the tail ($z > 0$) see a different RF phase than those at the head ($z < 0$).

2.3.2 Dispersive section

A magnetic chicane introduces a position shift that depends on energy:

$$\boxed{\begin{aligned} z_f &= z_i + R_{56} \delta_i + T_{566} \delta_i^2 + U_{5666} \delta_i^3 + \dots, \\ E_f &= E_i, \end{aligned}} \quad (2)$$

where $\delta_i = (E_i - E_0)/E_0$ is the relative energy deviation with respect to the chicane reference energy E_0 . The chicane is non-accelerating, so the energy is unchanged.

2.4 The state vector and why it is called R_{56}

In accelerator physics, the single-particle state is described by the 6D **state vector**:

$$\vec{X} = \begin{pmatrix} x \\ x' \\ y \\ y' \\ z \\ \delta \end{pmatrix} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \end{pmatrix},$$

where (x, x') and (y, y') are the transverse coordinates and angles, and (z, δ) are the longitudinal coordinates defined above.

To first order, a beam-transport element maps input to output via a 6×6 **transfer matrix** $\mathbf{R}$:

$$\vec{X}_{\text{out}} = \mathbf{R} \vec{X}_{\text{in}}. \quad (3)$$

The element R_{ij} gives the first-order dependence of the i -th output coordinate on the j -th input coordinate. In particular:

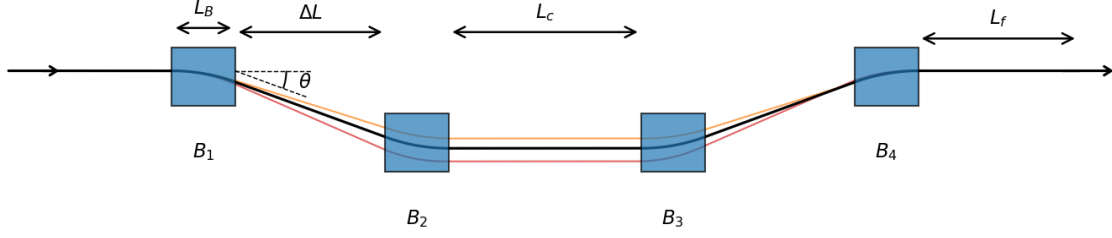
$$\boxed{\Delta z = R_{56} \delta}, \quad (4)$$

i.e., R_{56} is literally “row 5, column 6” of the transfer matrix — the first-order path-length dependence on energy deviation.

Higher-order effects are described by the tensors T_{ijk} , U_{ijkl} , etc.:

$$X_{i,\text{out}} = \sum_j R_{ij} X_{j,\text{in}} + \sum_{j \leq k} T_{ijk} X_{j,\text{in}} X_{k,\text{in}} + \dots$$

So T_{566} is the second-order coefficient coupling X_5 (output z) to $X_6 \cdot X_6$ (i.e., δ^2), and U_{5666} is the third-order term in δ^3 .



2.5 Chicane properties

For a symmetric four-dipole chicane with dipole length L_B , bend angle θ_0 , and drift length ΔL between adjacent dipoles, the dispersive coefficients are:

$$R_{56} = -2\theta_0^2 \left(\Delta L + \frac{2}{3} L_B \right), \quad (5)$$

$$T_{566} = -\frac{3}{2} R_{56}, \quad (6)$$

$$U_{5666} = 2 R_{56}. \quad (7)$$

Physical interpretation: A higher-energy particle ($\delta > 0$) bends less in each dipole, takes a shorter detour through the chicane, and arrives earlier ($\Delta z < 0$, toward the head). Hence $R_{56} < 0$ in the ELEGANT convention.

With $R_{56} < 0$,

Electrons with higher energies than reference particle travels a shorter path.

Electrons with lower energies than reference particle travels a longer path.

2.6 Compression condition

For a linearly chirped bunch ($\delta = a z_i$ where a is the chirp parameter set by off-crest RF), substituting into the chicane map:

$$z_f = z_i + R_{56} a z_i = z_i (1 + a R_{56}).$$

Full compression ($z_f = 0$ for all particles) requires:

$$a R_{56} = -1. \quad (8)$$

Since $R_{56} < 0$ in our convention, this requires $a > 0$: the bunch tail ($z > 0$) must have higher energy. This is achieved by accelerating the beam on the *falling slope* of the RF (behind the crest in the cosine convention, $\varphi_s > 0$).

2.7 RMS bunch length after compression

2.7.1 Chirp model

The off-crest RF imparts an energy–position correlation that is linear to leading order, with a quadratic correction from the RF curvature:

$$\delta_i = a z_i + b z_i^2, \quad (9)$$

where a is the linear chirp and b is the quadratic chirp (RF curvature plus any harmonic cavity contribution).

2.7.2 First-order compression

Substituting only the linear part of Eq. (9) into the chicane map (Eq. 2):

$$z_f = z_i + R_{56} a z_i = (1 + a R_{56}) z_i.$$

Taking the rms over the bunch:

$$\sigma_{z,f} = |1 + a R_{56}| \sigma_{z,i} = \frac{\sigma_{z,i}}{C}, \quad (10)$$

where the **compression factor** is:

$$C \equiv \frac{1}{|1 + a R_{56}|}. \quad (11)$$

Full compression ($C \rightarrow \infty$) corresponds to $a R_{56} = -1$, recovering the single-particle result (Eq. 8).

2.7.3 Second-order correction (T_{566})

Including both terms from Eq. (9) and the T_{566} coefficient in the chicane map:

$$z_f = z_i + R_{56}(a z_i + b z_i^2) + T_{566}(a z_i + b z_i^2)^2.$$

Expanding to second order in z_i (dropping $b^2 z_i^4$ and higher):

$$z_f = (1 + a R_{56}) z_i + (b R_{56} + T_{566} a^2) z_i^2. \quad (12)$$

To compute $\sigma_{z,f}^2 = \langle z_f^2 \rangle - \langle z_f \rangle^2$, we need the moments of the initial distribution. Writing $\alpha = 1 + a R_{56}$ and $\beta_2 = b R_{56} + T_{566} a^2$:

$$z_f = \alpha z_i + \beta_2 z_i^2.$$

The mean is:

$$\langle z_f \rangle = \beta_2 \sigma_{z,i}^2,$$

(representing a centroid shift, not compression). The variance:

$$\sigma_{z,f}^2 = \alpha^2 \sigma_{z,i}^2 + 2\alpha \beta_2 \langle z_i^3 \rangle + \beta_2^2 (\langle z_i^4 \rangle - \sigma_{z,i}^4).$$

For a **Gaussian** distribution, $\langle z_i^3 \rangle = 0$ and $\langle z_i^4 \rangle = 3\sigma_{z,i}^4$, so:

$$\sigma_{z,f}^2 = (1 + a R_{56})^2 \sigma_{z,i}^2 + 2(b R_{56} + T_{566} a^2)^2 \sigma_{z,i}^4. \quad (13)$$

The second term is always positive: the combined second-order effect (RF curvature + chicane nonlinearity) always increases the bunch length.

2.7.4 Minimum bunch length

At full compression ($aR_{56} = -1$), the first term vanishes and the minimum rms bunch length is:

$$\sigma_{z,\min} = \sqrt{2} |bR_{56} + T_{566} a^2| \sigma_{z,i}^2. \quad (14)$$

Using $T_{566} = -\frac{3}{2}R_{56}$ (Eq. 6) and $a = -1/R_{56}$ at full compression:

$$T_{566} a^2 = -\frac{3}{2}R_{56} \cdot \frac{1}{R_{56}^2} = -\frac{3}{2R_{56}}.$$

So the nonlinear coefficient evaluates to:

$$bR_{56} + T_{566} a^2 = bR_{56} - \frac{3}{2R_{56}}.$$

Without a harmonic linearizer (b from RF curvature alone), this sets the minimum achievable bunch length. The harmonic linearizer described below provides a means to choose b such that this coefficient vanishes.

2.8 Harmonic linearizer

The T_{566} nonlinearity can be compensated by adding a **harmonic RF cavity** (at the n -th harmonic of the main linac frequency) upstream of the chicane. The idea is to introduce a controlled quadratic curvature in the energy chirp that exactly cancels the T_{566} -induced distortion.

2.8.1 Energy chirp through both cavities

From the thin-lens map (Eq. 1), the main linac (V_1 , wavenumber k , phase φ_s) imparts:

$$\delta_1 = \frac{eV_1}{E_0} [\cos(kz_i + \varphi_s) - \cos \varphi_s],$$

where we subtract the reference-particle gain. Expanding to second order in kz_i :

$$\delta_1 = -\frac{eV_1 k}{E_0} \sin \varphi_s \cdot z_i - \frac{eV_1 k^2}{2E_0} \cos \varphi_s \cdot z_i^2 + \dots$$

The harmonic cavity (V_n , wavenumber nk , phase φ_n) similarly gives:

$$\delta_n = -\frac{eV_n nk}{E_0} \sin \varphi_n \cdot z_i - \frac{eV_n n^2 k^2}{2E_0} \cos \varphi_n \cdot z_i^2 + \dots$$

In practice the harmonic cavity is operated **on-crest** ($\varphi_n = 0$ or π), which sets $\sin \varphi_n = 0$ — eliminating any linear contribution — while maximising the quadratic term ($|\cos \varphi_n| = 1$). The total energy deviation is then:

$$\delta_i = a z_i + b z_i^2, \quad (15)$$

with:

$$a = -\frac{eV_1 k}{E_0} \sin \varphi_s, \quad b = -\frac{eV_1 k^2}{2E_0} \cos \varphi_s - \frac{eV_n n^2 k^2}{2E_0} \cos \varphi_n. \quad (16)$$

The linear chirp a is set entirely by the main linac; the quadratic curvature b receives contributions from both cavities.

2.8.2 Cancellation condition

From Eq. (12), the second-order distortion in z_f vanishes when:

$$\boxed{b R_{56} + T_{566} a^2 = 0 \implies b = -\frac{T_{566}}{R_{56}} a^2 = \frac{3}{2} a^2,} \quad (17)$$

where the last step uses $T_{566} = -\frac{3}{2}R_{56}$ (Eq. 6).

2.8.3 Solving for the harmonic cavity voltage

Substituting Eq. (16) into the cancellation condition (Eq. 17) with $\varphi_n = \pi$ (decelerating crest, so $\cos \varphi_n = -1$):

$$-\frac{eV_1 k^2}{2E_0} \cos \varphi_s + \frac{eV_n n^2 k^2}{2E_0} = \frac{3}{2} \frac{e^2 V_1^2 k^2}{E_0^2} \sin^2 \varphi_s.$$

Dividing through by $eV_1 k^2 / (2E_0)$:

$$-\cos \varphi_s + n^2 \frac{V_n}{V_1} = \frac{3eV_1}{E_0} \sin^2 \varphi_s.$$

Solving:

$$\boxed{\frac{V_n}{V_1} = \frac{1}{n^2} \left[\cos \varphi_s + \frac{3eV_1}{E_0} \sin^2 \varphi_s \right].} \quad (18)$$

In the high-energy limit ($eV_1/E_0 \ll 1$), the second term is negligible and:

$$\boxed{\frac{V_n}{V_1} \approx \frac{\cos \varphi_s}{n^2}.} \quad (19)$$

For example, with a third-harmonic cavity ($n = 3$) and $\varphi_s \approx 15^\circ$, this gives $V_3/V_1 \approx \cos 15^\circ / 9 \approx 0.11$. Setting $V_n = 0$ (no harmonic cavity) recovers the uncompensated case, and the minimum bunch length is limited by the T_{566} nonlinearity (Eq. 14).

2.9 Note on sign conventions

Different codes use different sign conventions for z and R_{56} :

Convention	Bunch head	R_{56} sign (chicane)	Codes
ELEGANT	$z < 0$	$R_{56} < 0$	ELEGANT, BMAD
MAD-X	$z > 0$	$R_{56} > 0$	MAD-X, PLACET

The physics is identical; only the sign of z is flipped. In the MAD-X convention, $z > 0$ means the particle is *ahead*, and a chicane has $R_{56} > 0$ (a higher-energy particle's position becomes more positive, i.e., moves toward the head). The relations $T_{566} = -\frac{3}{2}R_{56}$ and $U_{5666} = 2R_{56}$ hold in both conventions.

2.10 References

1. John Byrd, *Bunch compressor*, USPAS 2010.
2. Marco Venturini, *Single-Particle Longitudinal Dynamics and Magnetic Bunch Compression*, USPAS 2015.
3. Simone Di Mitri, *Bunch length compressors*, Cern Accelerator School 2016.

Chapter 3

Beam Breakup Instability

3.1 Introduction

In a linear accelerator, a charged beam excites electromagnetic wakefields as it passes through accelerating structures. The transverse component of these wakefields — the **dipole wake** $W_{\perp}$ — deflects trailing particles. If the wake-driven kick is resonant with the betatron oscillation of the trailing particles, the transverse amplitude grows without bound along the linac. This is the **beam breakup (BBU) instability**.

The essential physics is that of a driven oscillator at resonance: the leading part of the bunch oscillates freely and acts as the “drive,” while the trailing part responds. When drive frequency matches natural frequency, secular (linearly growing) solutions appear.

We develop the simplest model that captures this effect — the **two-macro-particle model** — derive the exact solution, identify the resonant growth, and show how **BNS damping** (Balakin–Novokhatsky–Smirnov, 1983) cures the instability.

3.2 Two-Macro-Particle Model

3.2.1 Setup and assumptions

We model the beam as two macro-particles, each carrying charge $eN/2$, where N is the total number of particles in the bunch. The two particles travel at the speed of light ($v = c$) through a **smooth-focusing channel** with (possibly different) betatron wavenumbers k_1 and k_2 .

The particles are separated longitudinally by a distance ℓ , typically taken as twice the rms bunch length: $\ell = 2\sigma_z$. The leading particle (particle 1) sees no wake from the trailing particle (causality), but the trailing particle (particle 2) experiences the transverse wake excited by the leading particle.

3.2.2 Wake coupling strength

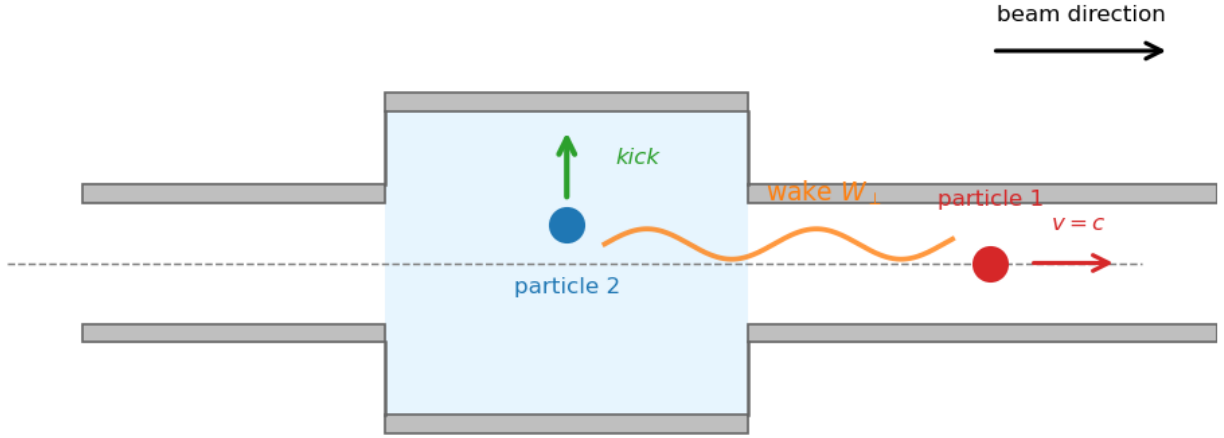
The transverse wake force on the trailing particle due to the leading particle’s offset x_1 is:

$$F_{\perp} = \frac{e^2 N}{2} W_{\perp}(\ell) x_1$$

where $W_{\perp}(\ell)$ is the transverse wake function evaluated at the separation ℓ (units: V/C/m²).

Dividing by the beam energy E (to convert force to the equivalent focusing term in the equation of motion for angle $x' = dx/ds$), we define the dimensionless coupling parameter:

$$G = \frac{e^2 N W_{\perp}(\ell)}{2E}$$



3.3 Equations of Motion

Using the longitudinal coordinate s as the independent variable (primes denote d/ds):

Leading particle (free betatron oscillation):

$$x_1'' + k_1^2 x_1 = 0 \quad (1)$$

Trailing particle (driven by the wake of particle 1):

$$x_2'' + k_2^2 x_2 = G x_1 \quad (2)$$

The structure is clear: eq. (1) is a free harmonic oscillator; eq. (2) is a **driven harmonic oscillator** where the drive comes from the transverse wake.

3.4 Solution of the Leading Particle

Equation (1) has the general solution:

$$x_1(s) = A_1 \sin k_1 s + B_1 \cos k_1 s$$

We choose initial conditions $x_1(0) = 0$ and $x_1'(0) = k_1 \hat{x}_1$ (i.e., the particle starts on axis with slope $k_1 \hat{x}_1$, so that $\hat{x}_1$ is the oscillation amplitude). Then $B_1 = 0$, $A_1 = \hat{x}_1$, and:

$$\boxed{x_1(s) = \hat{x}_1 \sin k_1 s} \quad (3)$$

3.5 Solution of the Trailing Particle (Detailed Derivation)

3.5.1 Step 1: The driven oscillator equation

Substituting eq. (3) into eq. (2):

$$x_2'' + k_2^2 x_2 = G \hat{x}_1 \sin k_1 s \quad (4)$$

This is a second-order linear ODE with constant coefficients, driven by a sinusoidal forcing term at frequency k_1 . We solve it by the standard method: general solution = homogeneous solution + particular solution.

3.5.2 Step 2: Homogeneous solution

The homogeneous equation $x_2'' + k_2^2 x_2 = 0$ has solution:

$$x_{2,h}(s) = A \sin k_2 s + B \cos k_2 s$$

3.5.3 Step 3: Particular solution (method of undetermined coefficients)

Since the driving term is $\sin k_1 s$ and $k_1 \neq k_2$ (we treat this case first; the degenerate case $k_1 = k_2$ is handled in the next section via a limit), we try:

$$x_{2,p}(s) = C \sin k_1 s$$

Differentiating twice:

$$x_{2,p}'' = -C k_1^2 \sin k_1 s$$

Substituting into eq. (4):

$$\begin{aligned} -C k_1^2 \sin k_1 s + k_2^2 C \sin k_1 s &= G \hat{x}_1 \sin k_1 s \\ C(k_2^2 - k_1^2) \sin k_1 s &= G \hat{x}_1 \sin k_1 s \end{aligned}$$

Solving for C :

$$C = \frac{G \hat{x}_1}{k_2^2 - k_1^2}$$

Therefore the particular solution is:

$$x_{2,p}(s) = \frac{G \hat{x}_1}{k_2^2 - k_1^2} \sin k_1 s \quad (5)$$

3.5.4 Step 4: General solution

The full solution is:

$$x_2(s) = A \sin k_2 s + B \cos k_2 s + \frac{G \hat{x}_1}{k_2^2 - k_1^2} \sin k_1 s$$

3.5.5 Step 5: Apply initial conditions

We impose the same initial conditions as the leading particle: $x_2(0) = 0$ and $x_2'(0) = k_1 \hat{x}_1$.

From $x_2(0) = 0$:

$$A \cdot 0 + B \cdot 1 + \frac{G \hat{x}_1}{k_2^2 - k_1^2} \cdot 0 = 0 \implies B = 0$$

From $x_2'(0) = k_1 \hat{x}_1$:

First compute $x_2'(s)$:

$$x_2'(s) = A k_2 \cos k_2 s + \frac{G \hat{x}_1 k_1}{k_2^2 - k_1^2} \cos k_1 s$$

Evaluating at $s = 0$:

$$A k_2 + \frac{G \hat{x}_1 k_1}{k_2^2 - k_1^2} = k_1 \hat{x}_1$$

Solving for A :

$$A = \frac{1}{k_2} \left(k_1 \hat{x}_1 - \frac{G \hat{x}_1 k_1}{k_2^2 - k_1^2} \right) = \frac{k_1 \hat{x}_1}{k_2} \left(1 - \frac{G}{k_2^2 - k_1^2} \right)$$

3.5.6 Step 6: Final result

Assembling the solution:

$$x_2(s) = \frac{k_1 \hat{x}_1}{k_2} \left(1 - \frac{G}{k_2^2 - k_1^2} \right) \sin k_2 s + \frac{G \hat{x}_1}{k_2^2 - k_1^2} \sin k_1 s \quad (6)$$

This can be rearranged into a more illuminating form by separating the “free oscillation” part from the “wake-driven” part:

$$x_2(s) = \frac{k_1}{k_2} \hat{x}_1 \sin k_2 s + \frac{G \hat{x}_1}{k_2^2 - k_1^2} \left(\sin k_1 s - \frac{k_1}{k_2} \sin k_2 s \right) \quad (7)$$

Verification: the first term is the free oscillation of particle 2 (with the same initial velocity as particle 1, hence the k_1/k_2 factor); the second term is the wake-driven correction that vanishes when $G \rightarrow 0$.

3.5.7 Verification by substitution

To confirm eq. (7) satisfies eq. (4), we compute $x_2'' + k_2^2 x_2$: - From the first term: $-k_1 k_2 \hat{x}_1 \sin k_2 s + k_2^2 \cdot \frac{k_1}{k_2} \hat{x}_1 \sin k_2 s = 0$ - From the $\sin k_1 s$ part of the second term: $\frac{G \hat{x}_1}{k_2^2 - k_1^2} (-k_1^2 + k_2^2) \sin k_1 s = G \hat{x}_1 \sin k_1 s$ - From the $\sin k_2 s$ part of the second term: $\frac{G \hat{x}_1}{k_2^2 - k_1^2} \cdot \frac{k_1}{k_2} (-k_2^2 + k_2^2) \sin k_2 s = 0$

The sum gives $G \hat{x}_1 \sin k_1 s$, which is the RHS of eq. (4).

3.6 The $\Delta k \rightarrow 0$ Limit: Beam Breakup

3.6.1 Taylor expansion

Now consider the critical case where both particles have nearly the same betatron wavenumber:

$$k_2 = k_1 + \Delta k, \quad |\Delta k| \ll k_1$$

We expand each factor in eq. (7) to first order in Δk .

The frequency difference:

$$k_2^2 - k_1^2 = (k_1 + \Delta k)^2 - k_1^2 = 2k_1 \Delta k + (\Delta k)^2 \approx 2k_1 \Delta k$$

The ratio k_1/k_2 :

$$\frac{k_1}{k_2} = \frac{k_1}{k_1 + \Delta k} \approx 1 - \frac{\Delta k}{k_1}$$

The sine function (this is the key expansion):

$$\sin k_2 s = \sin(k_1 s + \Delta k \cdot s)$$

Using the addition formula:

$$\sin k_2 s = \sin k_1 s \cos(\Delta k \cdot s) + \cos k_1 s \sin(\Delta k \cdot s)$$

For $\Delta k \cdot s$ not necessarily small in a global sense, but working to first order in Δk :

$$\sin k_2 s \approx \sin k_1 s + \Delta k \cdot s \cos k_1 s \quad (8)$$

3.6.2 Expanding the wake-driven term

Consider the bracket in eq. (7):

$$\sin k_1 s - \frac{k_1}{k_2} \sin k_2 s$$

Substituting our expansions:

$$\begin{aligned} & \sin k_1 s - \left(1 - \frac{\Delta k}{k_1}\right) (\sin k_1 s + \Delta k \cdot s \cos k_1 s) \\ &= \sin k_1 s - \sin k_1 s - \Delta k \cdot s \cos k_1 s + \frac{\Delta k}{k_1} \sin k_1 s + O(\Delta k^2) \\ &= -\Delta k \cdot s \cos k_1 s + \frac{\Delta k}{k_1} \sin k_1 s + O(\Delta k^2) \end{aligned}$$

Multiplying by $G\hat{x}_1/(k_2^2 - k_1^2) \approx G\hat{x}_1/(2k_1\Delta k)$:

$$\begin{aligned} & \frac{G\hat{x}_1}{2k_1\Delta k} \left(-\Delta k \cdot s \cos k_1 s + \frac{\Delta k}{k_1} \sin k_1 s \right) \\ &= -\frac{G\hat{x}_1}{2k_1} s \cos k_1 s + \frac{G\hat{x}_1}{2k_1^2} \sin k_1 s \end{aligned} \tag{9}$$

3.6.3 Expanding the free-oscillation term

$$\begin{aligned} & \frac{k_1}{k_2} \hat{x}_1 \sin k_2 s \approx \left(1 - \frac{\Delta k}{k_1}\right) \hat{x}_1 (\sin k_1 s + \Delta k \cdot s \cos k_1 s) \\ &= \hat{x}_1 \sin k_1 s + \hat{x}_1 \Delta k \cdot s \cos k_1 s - \frac{\Delta k}{k_1} \hat{x}_1 \sin k_1 s + O(\Delta k^2) \end{aligned} \tag{10}$$

3.6.4 Combining all terms

Adding eqs. (9) and (10):

$$x_2(s) = \hat{x}_1 \sin k_1 s + \hat{x}_1 \left(\Delta k - \frac{G}{2k_1} \right) s \cos k_1 s + O(\Delta k) \cdot \sin k_1 s$$

The $O(\Delta k) \cdot \sin k_1 s$ terms are bounded oscillations that do not grow with s . The dangerous term is:

$$\boxed{x_2(s) \approx \hat{x}_1 \sin k_1 s + \hat{x}_1 \left(\Delta k - \frac{G}{2k_1} \right) s \cos k_1 s} \tag{11}$$

3.6.5 Physical interpretation: secular growth = BBU

The term $s \cos k_1 s$ is a **secular term** — its envelope grows linearly with distance s . When $\Delta k = 0$ (identical focusing for both particles):

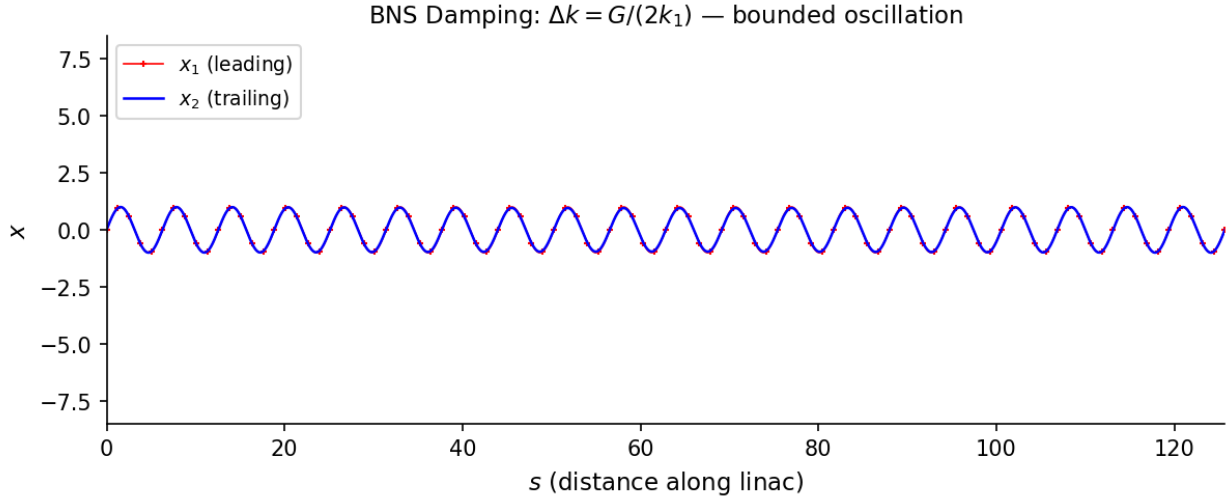
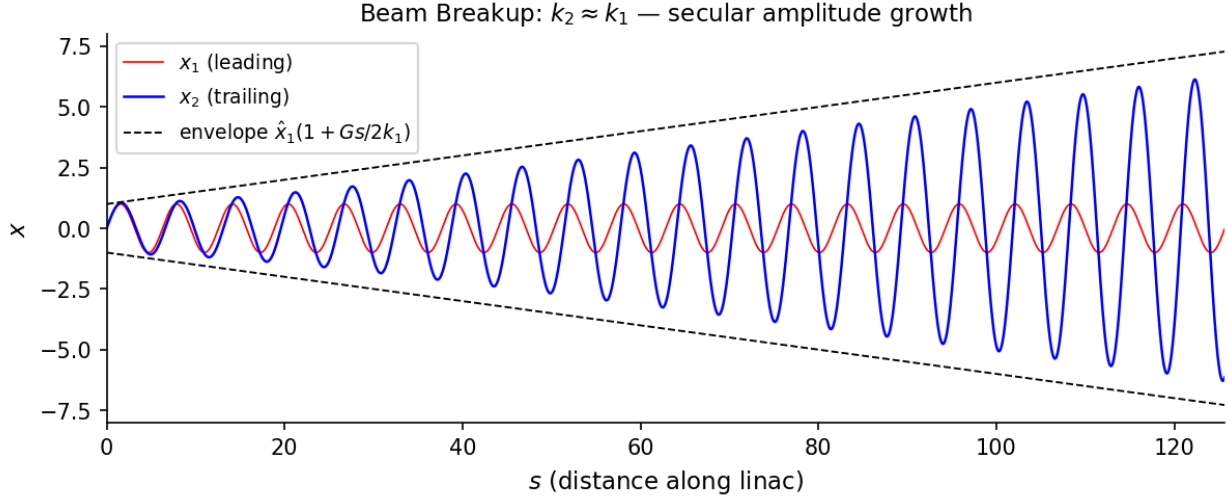
$$x_2(s) \approx \hat{x}_1 \sin k_1 s - \frac{G\hat{x}_1}{2k_1} s \cos k_1 s$$

The amplitude of the trailing particle grows as:

$$|x_2|_{\max} \sim \hat{x}_1 \left(1 + \frac{G}{2k_1} s \right)$$

This is **beam breakup**: the trailing particle's oscillation amplitude increases linearly with distance along the linac. In a long machine, this leads to beam loss on the beam pipe.

The physics is exactly that of a resonantly driven oscillator. The wake force from particle 1 oscillates at frequency k_1 . When particle 2 has the same natural frequency ($k_2 = k_1$), the drive is at resonance and the response grows without bound.



3.7 BNS Damping

3.7.1 The BNS condition

Inspecting eq. (11), the secular growth vanishes if we choose:

$$\Delta k = \frac{G}{2k_1} \quad (12)$$

This is the **BNS damping condition** (Balakin, Novokhatsky, and Smirnov, 1983).

When this condition is satisfied, the trailing particle still oscillates, but with a bounded amplitude — there is no secular growth.

3.7.2 Physical interpretation

The BNS condition requires that the trailing particle have a **slightly higher** betatron wavenumber than the leading particle: $k_2 = k_1 + G/(2k_1) > k_1$.

Physically, the extra focusing on the trailing particle provides a restoring force that exactly compensates the coherent wake kick. The trailing particle oscillates slightly faster than the leading particle, so it periodically slips in and out of phase with the drive, preventing the resonant build-up.

An equivalent viewpoint: the wake acts as an effective defocusing for the trailing particle. BNS damping adds extra focusing (via $\Delta k > 0$) to cancel this effective defocusing.

3.7.3 Practical realization

In a real linac, how does one give the trailing particle a different k_2 ?

Energy–chromaticity method: In a linac with magnetic focusing (quadrupoles), the betatron wavenumber depends on energy:

$$k \propto \frac{1}{\sqrt{p}} \propto \frac{1}{\sqrt{\beta\gamma}}$$

If the trailing particle has **lower energy** than the leading particle (by placing the bunch behind the RF crest), it experiences stronger focusing — hence higher k .

This is achieved by introducing a **correlated energy spread** $\Delta E/E$ along the bunch via RF phasing. The head of the bunch is at higher energy, the tail at lower energy. The natural chromaticity of the quadrupole lattice then provides the required Δk .

Quantitative estimate: With $k \propto p^{-1/2}$, we have $\Delta k/k \approx -\frac{1}{2}\Delta p/p$. The BNS condition becomes:

$$-\frac{1}{2} \frac{\Delta p}{p} = \frac{G}{2k_1^2}$$

$$\frac{\Delta p}{p} = -\frac{G}{k_1^2}$$

The required energy spread is proportional to the wake strength and inversely proportional to the focusing strength squared.

3.8 Summary

Quantity	Expression
Wake coupling	$G = e^2 N W_{\perp}(\ell)/(2E)$
BBU growth rate	$\sim G/(2k_1)$ per unit length
BNS condition	$\Delta k = G/(2k_1)$
Required $\Delta p/p$	$-G/k_1^2$

The two-macro-particle model reveals the essential mechanism: **resonant wake driving leads to linear amplitude growth**. BNS damping breaks the resonance by detuning the trailing particle’s betatron frequency just enough to compensate the wake-induced effective defocusing.

3.9 References

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